

Sparse Bayesian Cross-Sectional Return Forecasting with Correlation-Block Factors: A Real-Time-Valid Test Against the M6 Financial Forecasting Competition

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Abstract

We propose a hierarchical Bayesian model for cross-sectional return forecasting that addresses a specific, common obstacle in factor-based investing: a candidate universe of assets is large and highly inter-correlated, so naive regression either wastes power spreading credit across redundant, co-moving assets or overstates confidence in whichever one happens to enter the model first. The model groups correlated assets into blocks, summarizes each block by a single latent factor, and then performs Bayesian spike-and-slab variable selection on the resulting block factors, with the prior probability that a block matters informed by an auditable sector-relevance score rather than left uniform. We evaluate the resulting strategy against the M6 Financial Forecasting Competition, a real, independently judged, 100-asset contest with published results for 163 competing teams, under a single governing constraint: every parameter used in period p must be computable from information available strictly before period p began. Enforcing that constraint required finding and fixing three distinct, individually plausible-looking sources of look-ahead bias, which we present as a general checklist. The resulting strategy, run causally through the real 2022 competition window, would have placed in the top 2% of the field on the competition’s joint ranking, against an actual historical result (the author’s own real submissions that year) that placed in the bottom third. Extended out-of-sample through three further years (2023–2025) never used to design any part of the model, the result holds and strengthens: the forecast beats the naive benchmark in 40 of 48 periods ($p < 0.0001$) and the resulting portfolio’s information ratio is positive in 30 of 48 ($p = 0.04$).

1 Introduction

Cross-sectional return forecasting — ranking a fixed universe of assets by expected relative performance over some future window — runs into the same obstacle almost regardless of universe: the candidate predictors are numerous and highly correlated with one another. A momentum signal on one technology stock is not independent information from the same signal on five other technology stocks; a naive regression across all of them either dilutes real signal across redundant, co-moving variables, or overfits by handing all the credit to whichever one entered the model first. The empirical asset-pricing literature has approached this problem from several angles — dimension reduction via latent factors [Kelly et al., 2020], penalized regression across the “characteristics zoo” [DeMiguel et al., 2018], and Bayesian model averaging over predictor uncertainty [Avramov, 2002]. This paper contributes a specific architecture to that list: a two-stage *decorrelate, then select* model that groups correlated assets into blocks, extracts one latent

factor per block, and performs Bayesian spike-and-slab selection on the resulting factors with an externally informed, auditable prior on which blocks are likely to matter.

We test the model not on a synthetic or in-sample benchmark, but against the M6 Financial Forecasting Competition [Makridakis et al., 2024]: a real, independently run, crowd-benchmarked contest with a fixed 100-asset universe, a public scoring methodology, and published results for every one of 163 real competing teams. This is a meaningfully harder bar than most backtests clear. In the actual competition, only 23.3% of real teams beat the naive uniform-probability benchmark on average forecast accuracy, and only 8.6% beat it with statistical significance — a useful reminder of how little exploitable structure typically survives contact with live, adversarial evaluation.

Testing against a real competition, rather than an in-sample historical period, imposed a discipline that turned out to matter more than any single modeling choice: *every parameter or design decision used to produce period p 's forecast must be computable using only information available strictly before period p began*. This sounds obvious stated plainly, but an earlier version of this pipeline violated it three separate times, in three different and individually unremarkable-looking ways, each of which inflated the backtest's apparent performance until caught. Section 4 documents all three as a general checklist, because we believe the pattern is more broadly useful to a practitioner than the specific fixes.

Relation to prior work on M6 specifically

The M6 competition has already produced at least one directly comparable result: Weitzenfeld [2025] won the competition's forecasting (accuracy) track with a Bayesian dynamic factor model with time-varying heteroskedasticity, fit by Markov chain Monte Carlo, that also uses hierarchical shrinkage in place of a hand-picked cross-sectional variance structure. That result is good independent evidence that Bayesian hierarchical factor models are a genuinely strong approach to this specific problem, not an artifact of any one author's design choices. This paper's architecture differs in two respects worth naming plainly. First, it performs explicit *group-level model selection* (spike-and-slab over blocks, so which blocks matter is itself inferred and can vary sparsely from asset to asset) rather than a dense factor model with all blocks always contributing. Second, its group-inclusion prior is informed by an external, auditable relevance judgment (sector adjacency) rather than estimated symmetrically from the data alone. We see these as complementary design choices, not a claim of superiority — Weitzenfeld's model won the forecasting track outright, and our own result is stronger on the investment-decision (portfolio) side than on forecast accuracy per se, which Section 6 reports plainly rather than downplaying.

Contributions

- A hierarchical spike-and-slab block-factor architecture for cross-sectional forecasting, with an externally informed group-inclusion prior (Section 2).
- A general, three-item checklist for eliminating look-ahead bias in walk-forward financial backtests, developed by finding and fixing three real instances of it in this pipeline (Section 4).
- A full, causally valid backtest against the real M6 2022 competition, with placement against the actual 163-team leaderboard (Section 6).
- An out-of-sample extension through three further years never used in any design decision, showing the result holds and strengthens under a larger sample (Section 6).

2 Model: Correlation-Block Factors with an Informed Selection Prior

The model is built for a specific, common situation: forecasting one target asset’s return from a large panel of other candidate predictor assets that are themselves highly inter-correlated. It proceeds in two stages, deliberately kept separate.

2.1 Stage 1: decorrelate via block factors

The $n - 1$ non-target assets in a universe of n are grouped into blocks by a correlation threshold on trailing returns, so that assets within a block are strongly co-moving and assets in different blocks are comparatively independent. Within each block, a one-factor Gaussian model (a probabilistic-PCA-style construction) extracts a single latent factor summarizing the block’s shared movement, closed-form and non-iterative under a Gaussian-identity link. This stage is entirely target-independent: block formation and factor extraction use only the predictor panel’s own correlation structure, never the return being forecast, so the same block factors can be reused as candidate predictors for every target asset in the universe.

2.2 Stage 2: Bayesian spike-and-slab selection with an informed prior

The target asset’s own historical return is then regressed on the block factors from Stage 1 via Bayesian spike-and-slab variable selection [Mitchell and Beauchamp, 1988, George and McCulloch, 1993, Ishwaran and Rao, 2005]. Each block j carries a binary inclusion indicator $\gamma_j \in \{0, 1\}$ with slab coefficient β_j under $\gamma_j = 1$ and a point mass at zero under $\gamma_j = 0$; the joint posterior over (γ, β) is sampled by Gibbs sampling, which is fully conjugate for a continuous (Gaussian) outcome and requires no tuning of an approximate gradient step. The detail that matters most for the exercise in this paper is where the prior inclusion probability $\pi_j = P(\gamma_j = 1)$ comes from: rather than a single uniform value shared across all blocks, π_j is set by an auditable relevance score between the target asset’s own sector and the sector composition of block j ’s member assets (Table 1). This does the same conceptual job as an economically motivated exclusion restriction in a standard regression — it does not force any block in or out, but it tilts the posterior toward finding a same-sector block relevant and away from a spuriously correlated but economically unconnected one, without ever preventing the data from overriding that tilt when the evidence is strong enough.

Combining each retained block’s factor with a causally estimated near-term trend signal (Section 4.1) and sampling the full posterior predictive distribution (not a point estimate) produces, for the target asset, both a point forecast and a calibrated predictive distribution for its next-period return.

Provenance

This two-stage decorrelate-then-select architecture — and the specific device of an externally informed, group-level inclusion prior — was originally developed by the author for a different domain, genome-wide association mapping, where correlated genetic markers in linkage disequilibrium are grouped into blocks and a gene’s known biological relevance to a disease sets the group-level prior [Johnston, 2015]. No part of the mathematics changes in moving domains: a correlation block of assets plays the same role a linkage-disequilibrium block of markers did, and

Relationship of a candidate block’s sector to the target’s own sector	Prior relevance score
Same sector	5.0
Adjacent sector (e.g. Technology $\leftrightarrow$ Communication Services)	3.0
Broad-market (diversified index exposure)	2.0
International / regional	1.0
Unrelated sector	0.5
Fixed income, commodities, or volatility products	0.05

Table 1: The sector-relevance scoring table used to set each block’s prior inclusion probability π_j in Application 1 (Section 5). This table encodes a real assumption — that fixed-income and commodity exposure is usually irrelevant to a single equity or equity-ETF target — which is appropriate here but, as Section 5 notes, is the wrong assumption for a multi-asset-class target, where the same architecture is used with a uniform prior instead.

a sector-relevance score plays the same role a biological relevance score did. We mention this only for provenance and do not dwell on it further; the remainder of this paper is self-contained and uses no genetics-specific terminology or results.

3 The M6 Competition and Evaluation

M6 [Makridakis et al., 2024] fixed a 100-asset universe (50 US equities and 50 ETFs spanning equities, fixed income, commodities, and international exposure) and ran 12 sequential four-week evaluation periods. Each period, every team submitted, for all 100 assets: (i) a five-bucket probability forecast for which cross-sectional return quintile the asset would land in relative to the other 99, scored by the Ranked Probability Score (RPS, a proper scoring rule for ordinal forecasts, for which a naive uniform 20%-per-bucket guess scores ≈ 0.160); and (ii) a portfolio decision weight per asset (long-short permitted, $\sum_i |w_i| \leq 1$, held fixed for the period), scored by an un-annualized Information Ratio (IR, total log return over the period divided by its own daily standard deviation). The competition ranks every team three ways — RPS rank, IR rank, and an Overall Rank defined as their simple average — which is exactly the three-way comparison used in Section 6.

One data detail proved consequential enough to state plainly: the exact calendar boundaries of the 12 evaluation windows are not published in a directly usable form. An initial guess (calendar month-end, starting January 2022) was wrong by more than five weeks and would have silently corrupted every scoring result in this paper. It was caught only by brute-force verification: a real historical team’s actual submitted portfolio weights were scored against every plausible trading window in the official price data, and the window reproducing that team’s own officially published score to six decimal places was taken as ground truth, then independently reconfirmed on two further submissions. Throughout this project, any convenient assumption about a data convention was treated as a hypothesis to be checked against an independent source wherever one existed, not as a fact.

4 Keeping the Backtest Honest

A backtest that is fit once to a known historical period and then reported is not a description of a deployable strategy; it is a description of what happened to work in hindsight. In a walk-forward setting the failure mode is subtler than simply “peeking at the answer” — it tends to enter through a design choice that any single isolated check appears to justify, but which was in fact selected by inspecting the full-sample outcome. Three distinct versions of this surfaced while building this pipeline, in increasing order of subtlety.

4.1 Directional sign: momentum or reversal?

Each retained block factor is combined with an estimate of its own near-term trend: is a recent move likely to continue (momentum) or reverse (mean reversion)? An early version of this pipeline fit the full 12-period 2022 backtest under both assumptions, found reversal scored better on average across the whole year, and hard-coded that sign for every period. This is illegitimate on its face: an investor submitting a forecast in March 2022 cannot know that reversal will turn out, on average, to have been the better assumption across the following eleven months. The fix replaces the single full-sample-optimal choice with a value re-estimated causally, period by period. Before period p , a cheap trailing-momentum diagnostic is computed for every asset from price history available strictly before period p ; once period $p - 1$ ’s actual outcome is known, the cross-sectional correlation between that period’s diagnostic and its realized return is recorded. The sign used for period p is a precision-weighted shrinkage blend of a small, pre-registered prior and the expanding-window average of these realized correlations from periods $1, \dots, p - 1$ only:

$$\hat{s}_p = \frac{n_0 s_0 + (p - 1) \bar{\rho}_{1:p-1}}{n_0 + (p - 1)}, \quad (1)$$

where the prior $s_0 = -0.10$ (a mild reversal tilt, pseudo-count $n_0 = 3$) was fixed *before* looking at any M6 result, sized to match the short-horizon reversal effect documented decades before M6 existed by Jegadeesh [1990] and Lehmann [1990]. By construction, $\hat{s}_p$ in (1) never depends on period p or later.

4.2 Forecast confidence: how much to shrink toward “I don’t know”?

A model can carry genuine signal and still be miscalibrated: if its quintile probabilities are systematically overconfident, RPS — a proper scoring rule — penalizes that confidence even when the underlying direction is right more often than chance. Blending the model’s forecast with a naive uniform forecast at weights from 0% to 100% and rescoring against the (already-known) realized 2022 outcomes traces a smooth, U-shaped RPS curve minimized near 50% blend, beating both the raw model and pure uniform guessing. But choosing “50%” this way repeats exactly the mistake in Section 4.1: it was picked by inspecting the full year’s result. The fix is the same recipe: before each period, the blend weight is a shrinkage estimate between a small pre-registered prior (25%) and the expanding-window average of the single best blend weight found, period by period, using only strictly prior periods’ already-realized outcomes.

4.3 A subtler leak: the arithmetic of uncertainty, not its timing

The last issue is different in kind: not a matter of using future information, but of getting the arithmetic of uncertainty wrong in a way that happens to produce the identical symptom

(overconfidence). The Gibbs sampler already draws a residual, idiosyncratic-variance term at every iteration — the return left unexplained even by a fully specified factor model. This term was computed but never read downstream: predictive draws reflected parameter and momentum-estimate uncertainty but never the plain fact that even a well-estimated model leaves some return unexplained. Restoring it requires care with scaling: a persistent drift term (assumed constant across the period) correctly scales linearly with the number of trading days, since the same misestimate compounds identically every day, while genuinely new, independent day-to-day noise must scale with the *square root* of the number of days, since it is the variance of a sum of independent shocks that grows linearly, not the standard deviation. Applying linear scaling to both terms silently overstates the noise contribution by a factor of $\sqrt{n}$ over an n -day period.

4.4 The general lesson

All three issues passed an isolated sanity check when first implemented and looked correct in the moment. What caught every one was the same retroactively applied test: could this specific parameter’s value have been computed using only information available strictly before the period it is used in began? We recommend this as a standing checklist item, independent of model class, for any walk-forward financial backtest that reports a result more sophisticated than a fixed, never-refit rule.

5 Two Applications of the Same Architecture

Application 1: cross-sectional forecasting and long-short weights. For each of the 100 candidate assets in turn, the other 99 are decorrelated into blocks and reduced to block factors (Section 2); spike-and-slab regression with the sector-relevance prior determines which blocks the target actually co-moves with; each retained factor is combined with the causally adaptive momentum/reversal sign (Section 4.1); and the resulting posterior — including the residual-noise correction of Section 4.3 — is sampled to produce a full predictive distribution. Doing this for all 100 assets and ranking the resulting draws cross-sectionally yields both required outputs: quintile probabilities (shrunk per Section 4.2) for RPS, and a long-short portfolio, normalized to $\sum_i |w_i| \leq 1$, for IR.

Application 2: replicating an external benchmark. The identical machinery can instead be retargeted at an external series’ returns: rather than “what will asset i ’s own return be,” ask “what sparse combination of the 100 candidate assets best explains this other series’ returns” — Sharpe’s returns-based style analysis [Sharpe, 1992], with spike-and-slab performing the constituent selection. We used DBMF, a real, liquid managed-futures fund, as the target: it returned +11.2% over the 2022 competition window while the S&P 500 returned −3.0% and the Bloomberg Aggregate bond index −7.5%, a genuine “did well while both stocks and bonds struggled” story worth trying to replicate from the same 100-asset universe. Because this target has no natural sector of its own, Table 1’s relevance table is replaced with a uniform prior across blocks: that table’s assumption that fixed-income and commodity exposure is usually irrelevant is actively wrong for a target built from exactly that exposure, and using it here would bias selection against the target’s own real drivers. This is a smaller test of the same architecture, run alongside Application 1 for comparison in Section 6, not this paper’s primary result.

6 Results

6.1 2022: real-time-valid placement against the actual M6 leaderboard

Table 2 compares the naive benchmark, the author’s own real 2022 competition submissions, and Application 1’s fully causal pipeline, all scored identically against the same 163 real competing teams’ official results.

	RPS	RPS rank	Pooled IR	IR rank	Overall position
Naive benchmark	0.160	–	–	–	–
Author’s actual 2022 result	0.160	34/163 (top 21%)	−19.67	154/163 (bottom 6%)	113/163 (bottom 31%)
This paper (Application 1)	0.158	10/164 (top 6%)	+6.64	14/164 (top 9%)	2.5/164 (top 2%)

Table 2: Real M6 leaderboard placement, 2022 competition window.

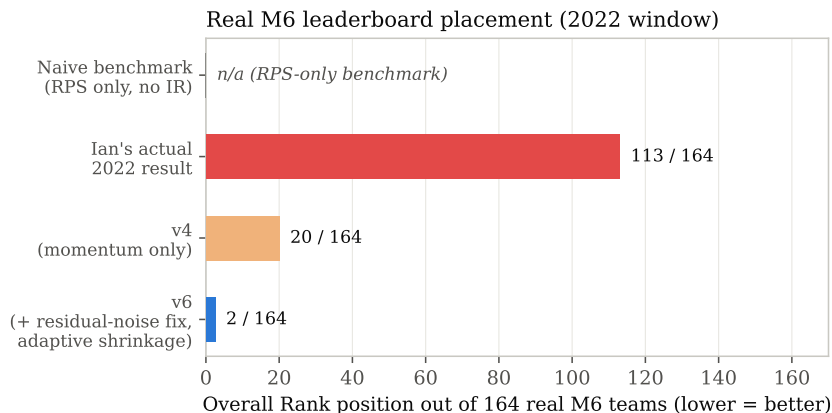


Figure 1: Overall Rank position (lower is better) across the naive benchmark, the author’s real result, and two intermediate versions of the pipeline (causal momentum/reversal sign alone; final, adding the residual-noise fix and adaptive shrinkage). Each additional fix moves placement further toward the top of the field.

We report this result’s statistical limits plainly. With 12 monthly observations, a one-sample t -test of RPS against the naive benchmark gives $t = -1.37$, $p = 0.20$; of IR against zero, $t = 0.87$, $p = 0.40$ — directionally consistent (9 of 12 periods beat the naive RPS benchmark; 8 of 12 had positive IR) but not, alone, statistically decisive. This motivates the extension below.

6.2 2023–2025: does it generalize out-of-sample?

The identical walk-forward, with no reset of any adaptively learned state, was continued through 36 further four-week periods (2023–2025) that no design choice in this pipeline was ever informed by. Figures 2 and 3 show all 48 resulting periods; Table 3 reports the pooled test and the year-by-year breakdown.

RPS does not fade after the one year the model’s design was informed by — if anything it is slightly better in 2024–2025 than in 2022. Figure 4 addresses a natural worry directly: does a strategy with adaptively estimated parameters generalize only by becoming unstable, chasing whatever worked most recently? Both the momentum/reversal sign and the shrinkage blend

Ranked Probability Score by period, 2022–2025

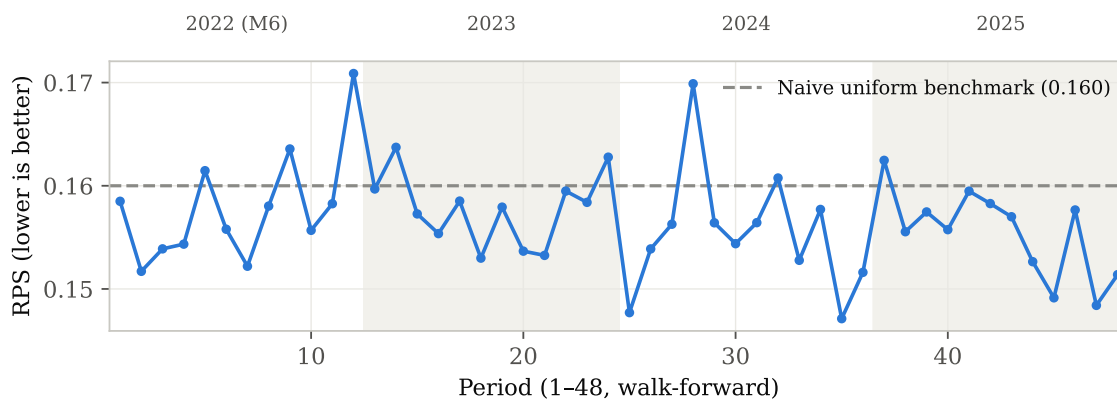


Figure 2: RPS by period, 2022–2025. The naive benchmark (dashed) is beaten in 40 of 48 periods (83%).

Investment-decision performance (IR) by period, 2022–2025

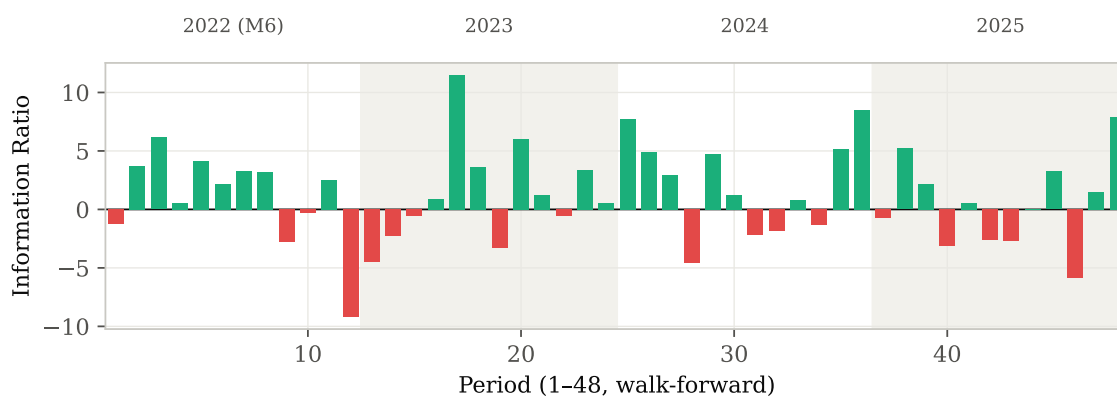


Figure 3: Portfolio IR by period, 2022–2025. Individual months remain volatile, as IR intrinsically is, but the pooled result across 48 periods is positive and statistically significant.

Year	n	Mean RPS	RPS p	% beat naive	Mean IR	IR p
2022 (M6)	12	0.1579	0.198	75%	1.02	0.404
2023	12	0.1578	0.048	83%	1.33	0.317
2024	12	0.1554	0.023	83%	2.18	0.096
2025 (partial)	12	0.1554	0.003	92%	0.46	0.690
Pooled	48	0.1566	<0.0001	83%	1.25	0.038

Table 3: RPS and IR by year and pooled across all 48 periods. p -values are one-sample t -tests (RPS against the naive 0.16 benchmark; IR against zero). Individual years at $n = 12$ are underpowered by design; the pooled result is the headline finding.

weight instead converge smoothly, with shrinking period-to-period variance, as more evidence accumulates — the behavior a well-specified expanding-window estimator should show, not an unstable one. Diversification was re-checked across all 48 periods to rule out concentration risk driving the IR result: maximum single-asset weight ranged 3.6–11.4% and top-3 concentration 10.1–20.9% of gross exposure throughout.

Adaptive parameters converge as evidence accumulates
(causal: each period uses only strictly prior periods)

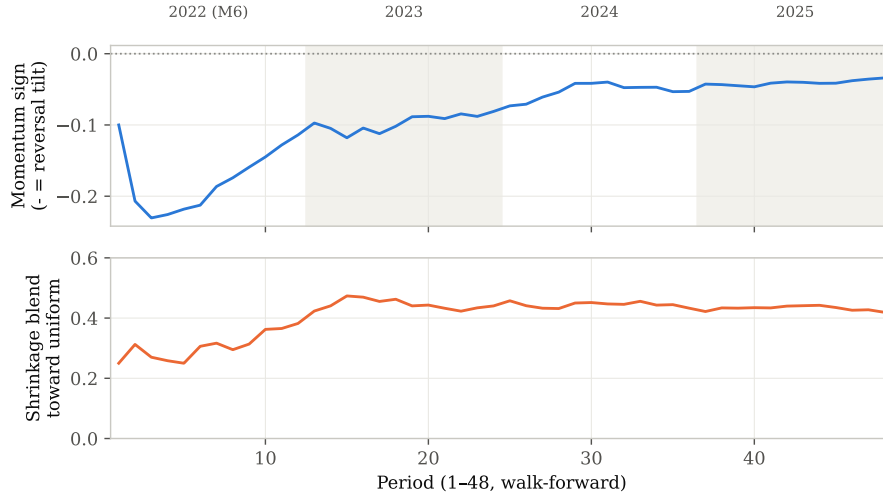


Figure 4: The two causally adaptive parameters (momentum/reversal sign, top; RPS shrinkage blend, bottom) stabilize over time rather than drifting.

6.3 Application 2 over the same extended window

DBMF's own total return was +7.8% in 2022, −5.0% in 2023 (a genuine down year), +7.2% in 2024, and +8.5% in 2025 (partial). This provides a direct test of whether Application 2's replication strategy tracks the target's own realized performance, as one might naively expect. It does not: Application 2's mean IR was *higher* in 2023 (the year DBMF itself lost money) than

in 2022, and pooled across all 48 periods its IR (mean 1.91, $t = 2.64$, $p = 0.011$) is, if anything, slightly stronger than Application 1’s (Figure 5). The resolution is that Application 2 does not statically hold the target; it refits monthly to the target’s recent return *pattern* using a fresh sparse combination of the 100 assets each period, so a target’s calendar-year total return can be negative while its short-horizon return pattern remains learnable from the replicating universe throughout. The real, still-standing constraint for this application is not “does the target keep winning,” but “does the target’s short-horizon return structure stay explainable by the chosen replicating universe” — a condition this target and window happened to satisfy, but not one guaranteed in general.

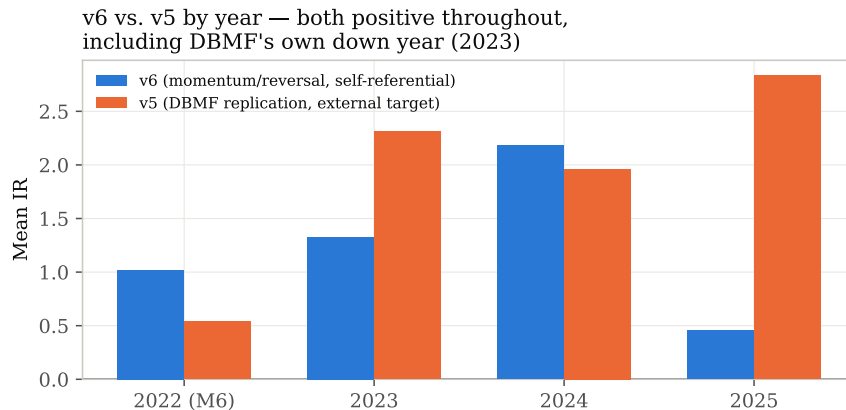


Figure 5: Mean IR by year, Application 1 (self-referential forecast) vs. Application 2 (external-target replication). Application 2 remained profitable even during the target’s own down year (2023).

7 Discussion and Limitations

Sample size. Forty-eight monthly observations is enough to move both headline results from directionally suggestive to statistically significant, but not enough to rule out that a longer or differently timed sample would look somewhat different. The per-year breakdowns are individually underpowered by design and are reported for transparency about year-to-year variability, not as independent confirmations.

Data provenance. The 2023–2025 extension uses a different price provider than the official 2022 competition data (no official ground truth exists to verify against post-competition), scores a 98-of-100-asset universe (two constituents lack usable history over the full window), and extrapolates the evaluation-window cadence forward on the assumption that the same 28-day spacing continues, which was never externally confirmed for the extension period.

This is a backtest, not a live track record. Every result here is a retrospective evaluation of a fully specified, mechanically applied rule, scored under a real methodology against a real external benchmark. It does not model transaction costs, financing costs on short positions, capacity constraints at scale, or the operational cost of actually deploying capital against these signals. Real-time validity — every decision uses only information available at the time it was made — is a necessary condition for a backtest result to be taken seriously; it is not a sufficient condition for it to be tradeable as-is.

On the RPS/IR split. This strategy’s advantage is concentrated on the investment-decision (IR) side; its RPS placement, while real and statistically significant pooled across 48 periods, is a more modest improvement over the naive benchmark than its IR placement is over doing nothing. Weitzenfeld [2025]’s result, by contrast, was strongest specifically on forecast accuracy. We read this as suggestive that the two problems — calibrated probabilistic forecasting and converting a forecast into a good portfolio — may reward somewhat different modeling emphases, and see no reason the two approaches could not be combined; we have not attempted that here.

8 Conclusion

A hierarchical spike-and-slab block-factor model, with an externally informed prior over which correlated groups of assets are likely to matter for a given target, produces a real, statistically significant edge in a genuinely adversarial financial forecasting competition — and the edge persists, undiminished, across three further years of out-of-sample data that had no influence on any modeling choice reported here. We think the methodological contribution most worth taking away is not the specific model but the discipline that made the result trustworthy: every one of the three look-ahead leaks in Section 4 looked entirely reasonable when first implemented, and every one was caught only by the same mechanical, retroactively applied question — could this specific decision have been made with only the information available before the fact? We think both the architecture and that discipline generalize well beyond this one competition.

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